

Scanned-page crop tightening + image→3D scaffold (v0.77.0)

Sharper figure crops on scanned manuals, and an opt-in (clearly-labeled) local image→3D experiment.

Figure-crop boundary — now precise on scanned pages

❏ OCR word-box caption

- No text layer? OCR the page (pytesseract) to find
- the 'FIGURE n' caption, crop the illustration ABOVE it.
- Gated by VIEWER_FIGCROP_OCR; degrades if no Tesseract.

Precise, the requested path.

❏ Row ink-density fallback

- No OCR? numpy row-darkness profile finds where the
- dense parts-table band starts, crop above it.
- Pure CPU, no extra services.

Always-available fallback.

❏ Result

- Synthetic image-only page: cropped top 49% via the
- OCR caption (was a loose 62%).
- Born-digital pages unchanged. top-62% remains last resort.

Tighter, verified.

Local image→3D — EXPERIMENTAL, opt-in, never authoritative

Off by default. Watermarked 'ARTISTIC APPROXIMATION — NOT TO SCALE'. The authoritative image stays the Manual illustration (cited figure).

figure crop

the part's cited PNG



your GPU backend

{in}->{out} OBJ (TripoSR/InstantMesh/...)



mesh3d sidcar

OBJ cached; index never written



Approximation tab

loaded in WebGL, watermarked

Configure: VIEWER_IMG3D_CMD env or engine/image3d_backend.txt · GET/POST /api/image3d · GET /api/image3d_mesh · docs/IMAGE3D-SETUP.md

VERIFIED

Scanned crop: image-only synthetic -> OCR caption at y≈372 -> clip top 49%. image→3D scaffold end-to-end: unconfigured returns a setup pointer; a trivial cube backend -> generate runs -> OBJ parsed to V=4/F=2 -> mesh_url served. Routes added to RUN-ALL-TESTS.bat. Honest: AI 3D is an approximation, gated + watermarked + off by default.